THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS VIII — SCIENCE · MID-TERM PRACTICE PAPER

Session 2026-27 · Term I · Chapters 1–7

Class: 8	Subject: Science	Set 01	Time: 2 Hours 30 Minutes	Max. Marks: 60

General Instructions:

1. All questions are compulsory.
2. This question paper has five sections A, B, C, D and E. Marks are indicated against each question.
3. Section A (Q. 1, parts I–XII) consists of multiple choice questions of 1 mark each.
4. Section B (Q. 2–5) consists of very short answer questions of 2 marks each.
5. Section C (Q. 6–8) consists of short answer questions of 3 marks each.
6. Section D (Q. 9–11) consists of long answer questions of 5 marks each.
7. Section E (Q. 12–15) consists of four case-based / source-based questions of 4 marks each (1 + 1 + 2).
8. There is no overall choice. However, internal choice is given in some questions. Only one of the choices in such questions is to be attempted.
9. Draw neat and labelled diagrams wherever required.

SECTION A — Multiple Choice Questions — 1 questions · 12 marks

Question 1 has twelve parts (I–XII). Each carries 1 mark.

1. From Question I–XII, choose the correct option.

I. Two identical beakers have cold and hot water. Equal spoons of sugar are added to test where sugar dissolves faster. Which should be kept the SAME in both beakers? 1

(a) Speed of dissolving (b) Volume of water and amount of sugar
(c) Temperature of water (d) Time taken to dissolve

II. Which of the following is a unicellular protozoan? 1

(a) Penicillium (b) Amoeba (c) Spirogyra (d) Mucor

III. Rhizobium bacteria living in the root nodules of pulse plants help the plant by: 1

(a) causing root rot
(b) making curd
(c) fixing atmospheric nitrogen into usable form
(d) absorbing sunlight

IV. Which of these diseases can be prevented by vaccination? 1

(a) Cancer (b) Diabetes
(c) High blood pressure (d) Measles

V. Which statement about antibiotics is correct? 1

(a) They can be taken in any quantity for any illness
(b) They act on bacteria and should be taken only on a doctor's advice
(c) They are made only from viruses
(d) They cure viral infections such as the common cold

VI. Which change will NOT increase the strength of an electromagnet? 1

(a) Increasing the number of turns in the coil
(b) Replacing the iron core with a wooden rod
(c) Using a soft iron core
(d) Connecting one more cell in series

VII. A fuse wire is chosen because it has: 1

(a) a low melting point (b) a very high melting point
(c) no resistance (d) a very high conductivity only

VIII A football rolling on the ground gradually slows down and stops. The force mainly responsible is: 1

(a) electrostatic force (b) buoyant force

(c) magnetic force (d) friction

IX A force of 200 N acts perpendicularly on an area of 0.5 m². The pressure is: 1

(a) 100 Pa (b) 200 Pa (c) 0.0025 Pa (d) 400 Pa

X The wall of a dam is made thicker at the bottom because: 1

(a) pressure of water increases with depth (b) pressure decreases with depth

(c) water is heavier at the surface (d) water exerts no pressure sideways

XI A drop of ink spreads evenly in a glass of still water after some time. This shows that: 1

(a) particles of matter move and mix by themselves

(b) ink and water react chemically

(c) ink particles are heavier than water

(d) water particles are at rest

XII Which of the following is TRUE for particles of matter? 1

(a) They have no force of attraction

(b) They are completely at rest in solids

(c) They are always in motion and have spaces between them

(d) They are very large and visible to eye

SECTION B — Very Short Answer Questions — 4 questions · 8 marks

2 marks each (about 30–40 words).

2. Differentiate between an *observation* and an *inference* with one example. **2**

OR

What is a *variable* in an experiment? Give one example of a variable that is kept constant.

3. Why is it advised to boil drinking water in flood-affected areas? **2**

4. State any two differences between a permanent magnet and an electromagnet. **2**

5. A sharp knife cuts a vegetable easily but a blunt knife does not, though the same force is applied. Why? **2**

SECTION C — Short Answer Questions — 3 questions · 9 marks

3 marks each (about 60 words).

6. Name the three groups of microorganisms other than viruses and give one example of each. **3**

OR

Explain how milk is converted into curd.

7. Design a simple activity to show that liquids exert pressure on the walls of a container. **3**

8. Distinguish between solids, liquids and gases on the basis of arrangement of particles, inter-particle space and shape. **3**

OR

Perfume sprayed in one corner of a room can be smelt in the opposite corner after some time. Explain, using the particle idea.

SECTION D — Long Answer Questions — 3 questions · 15 marks

5 marks each.

9. Complete the given sentences (choose from the words in the box): **(Bacteria, Vaccine, Electromagnet, Pascal, Friction)** **5**

i. A _____ is a preparation that trains the body to fight a specific disease.

ii. The SI unit of pressure is the _____.

iii. A temporary magnet made by passing current through a coil wound over soft iron is a(n) _____.

iv. _____ opposes the relative motion between two surfaces in contact.

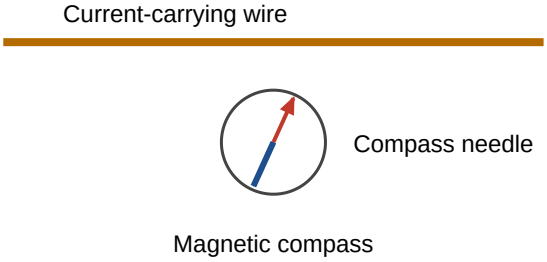
v. _____ are microorganisms that are single-celled and have no true nucleus.

10. Match the Columns **5**

Column I	Column II
(a) Barometer	(i) alloy of high resistance and high melting point
(b) Heating element	(ii) measures atmospheric pressure
(c) Penicillin	(iii) force with which Earth attracts a body, measured in newton
(d) Weight	(iv) mixing of particles of one substance into another on their own
(e) Diffusion	(v) first antibiotic obtained from a mould
	(vi) force that opposes motion

11. (a) Describe an activity to show that a current-carrying wire deflects a magnetic compass needle. (3) **5**

(b) What happens to the deflection when the direction of the current is reversed? (2)



Magnetic compass

OR

(a) Distinguish between mass and weight (any three points). (3)

(b) A body has mass 5 kg. Find its weight on Earth (g = 10 m/s²). (2)

SECTION E — Case-based / Source-based Questions — 4 questions · 16 marks

Read the cases carefully and answer. Each question carries 4 marks.

12. In Class VIII many students of Meena's class came to school with cough, running nose and fever. The teacher told them to stay at home, rest and drink plenty of fluids, and informed that the disease spreads from person to person.

(a) Name the type of disease that spreads from one person to another. 1

(b) Name one way through which such a disease can spread. 1

(c) Suggest two precautions to prevent the spread of this disease in the class. 2

OR

(d) Why will an antibiotic tablet not cure this illness if it is caused by a virus?

13. In the school science club, Arjun wound 40 turns of insulated copper wire around an iron nail and connected the ends to a 3 V cell through a switch. The nail picked up a few iron pins when the switch

was closed.

(a) What type of magnet was formed? 1

(b) What happens to the pins when the switch is opened? 1

(c) State any two ways by which Arjun can make the nail pick up more pins. 2

OR

(d) Why is insulated wire used to make the coil?

14. On a hot summer evening dark clouds gathered, strong winds blew and a thunderstorm with lightning occurred. Roof sheets of a few houses were blown off.

(a) What causes lightning? 1

(b) Why do high-speed winds blow off roofs? 1

(c) Mention two safety measures one should follow during a thunderstorm. 2

OR

(d) Why is it unsafe to stand under a tall tree during lightning?

15. A cricket ball rolled on a smooth cement pitch stopped after travelling 30 m. When the same ball was rolled with the same push on a grass field, it stopped after only 12 m.

(a) Name the force that stops the ball. 1

(b) Is friction a contact force or a non-contact force? 1

(c) Why does the ball stop earlier on grass than on cement? 2

OR

(d) Mention two ways in which friction can be reduced.